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A Highly Sensitive Self-Assembled-Nanoparticles e-Skin Sensor for Controlling Avatar Facial Micro-Expressions

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With their unique electrical, mechanical, and surface properties, gold nanoparticles (AuNPs) open up new possibilities for sensor technology. In particular, conductive thin films constructed from ligand-stabilized AuNPs are considered an ideal sensing platform due to their high surface area, excellent conductivity, and biocompatibility. However, most methods for making conductive AuNPs thin-film sensors with excellent sensitivity require expensive equipment. In this work, an innovative resistive strain sensor consisting of AuNPs and poly (allylamine hydrochloride) (PAH) based on the mutual adsorption of positive and negative charges using a low-cost layer-by-layer self-assembly (LBL-SA) approach on a flexible polyester substrate is developed. The conductance changes at low temperatures of the AuNPs/PAH agree with the Arrhenius-type activation of charge transport. Additionally, the maximum gauge factor of the sensor is shown experimentally to be ≈ 656 when 1% strain is applied to the sensor film. This work demonstrates that the sensor detects body motions, eyeball movements, and facial micro-expressions. For detecting eyeball movements and facial micro-expressions, the macro-recall can reach 91.5% and 98.8%. Simultaneously, the sensor can control the virtual avatar's eye movements and human facial micro-expressions in VR. Therefore, nanoparticle-based sensors can be extensively used in future applications related to healthcare and human-computer interaction.

1. Introduction

The application of ligand-stabilized nanoparticles offers interesting conceptions for strain sensors as it has been determined that nanomaterials can be used to construct sensors with better sensitivity and customizable adaptability.^[1,2] Due to their simplicity of fabrication, gold nanoparticles (AuNPs) have been extensively documented and adapted as a zero-dimensional nanomaterial embedded in conductive films,^[3] which can be used to develop flexible sensors with excellent sensitivity to sense multi-directional stresses on the skin of the human body.^[4,5] However, many challenges remain in the preparation of ligand-stabilized thin films of AuNPs. In particular, the influence of the interparticle distance and the structural properties of the stabilized ligands on charge transport.^[6] For example, Kotov and co-workers^[7] demonstrated the mechanism of charge transfer in AuNPs/polyurethane films, and they concluded that the charge dynamics in

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Table 1. A comparison of gauge factors, experimental methods, and applications of skin resistance strain sensor.

Nanomaterials	Methods	Gauge factor (Max.)	Applications
Carbon nanotube ^[27]	Self-assembly	62.3	Facial expressions and eye tracking
Polyurethane-PDMS ^[20]	Electrospinning	6.3	Facial skin strain detection
Au-nanowire ^[19]	Chemically bonded	≈82.5	Facial expressions monitoring
AuNPs/PDMS ^[18]	Template-induced printing	–	Recognition of human facial expression
Graphene ^[28]	Laser-induced process	41	Eye tracking
AuNPs/PAH (Our work)	Layer-by-layer self-assembly	656	Recognition of human facial expression, eye-ball movement tracking, and controlling avatar facial expressions

such nanocomposites can be described on the basis of the classical percolation theory; moreover, they described the possible rearrangements of the nanoparticles when the AuNPs/polymer system is under tensile or compressive stress. Besides, Ko^[8] and Kim,^[9] obtained similar charge transport processes in silver nanowire percolation network films and AuNPs network films. Collectively, these results suggest that charge transport in such composites occurs through an activated tunneling process. Thus, significant progress has been made in the application of nanoparticle-based components in flexible electronic skin.^[10]

The spacing between particles in metal nanoparticle polymer membranes is a key factor in determining the electrical conductivity of the membrane. However, only a few of them properly define the internal electrical sensing mechanism of the nanocomposite films containing AuNPs. For example, Vossmeier and co-workers^[11] prepared AuNPs/organic films via layer-by-layer self-assembly (LBL-SA) and showed linear current–voltage characteristics and Arrhenius-type of charge transport. They revealed the sensing mechanism and laid down the foundation for further quantitative understanding of the charge transfer mechanisms and the resistance sensing mechanisms of metal nanoparticles/organic composite films. Jia and co-workers^[12] showed conductive behaviors of carbon nanotube (CNT)/polydimethylsiloxane (PDMS) under tensile strain. They investigated the temperature-dependent resistivity and conductivity of CNT ribbons, revealing that the conductivity is governed by a 3D hopping process. A Monte Carlo model was used to simulate the flow of charge and the effect of disorder in 1D and 2D nanoparticle arrays on the conductivity of metal nanoparticles.^[13] The percolation theory is effectively supported by the experimental results of the AuNPs composite films, although an additional factor, accounting for the self-arrangement of the nanoparticles is required in the description.^[14] Notably, according to percolation theory, two charge transport models have been utilized to express the conductivity of the nanoparticle-based strain sensor:^[12,15] Specifically, they are 1) the tunneling mechanism model (TMM) governed by the different distances between the neighboring nanoparticle; 2) the variable range hopping (VRH) of charges due to the size of the nanoparticles and the electrical properties of the host matrix or their mutual action. Hence, e-skins based on AuNPs/polymer nanocomposites that support charge transfer mechanisms during deformation have been developed successfully already, and are gradually becoming one of the key components of new flexible electronic skin.^[16]

At present, flexible electronic skin has been extensively developed for human physiological signals^[17] and facial expression detection and recognition (Table 1).^[18] However, micro-expression recognition based on electronic skin is only limited to facial muscle detection,^[19] and there are very few applications of facial expression recognition and the use of expression recognition to judge a person's emotional state. In contrast, basic human facial expressions^[20] and eye movements^[21] are the main external expressions of human emotions. Meanwhile, one of the important potential applications faced by flexible e-skins is the combination with virtual technologies.^[22] Tracking human emotional states (through tracking facial expressions) and eye movements will enable advanced human-avatar interfaces for developing more realistic metaverse or virtual worlds in the future. In addition, eye movement data can also detect the reading order and habits,^[23] cybersickness,^[24] and eye vergence^[25] in virtual reality (VR) environments. With the development of related technology, the virtual world of the future will be more like a real society, with immersive experiences and more direct physical interaction.^[26] Here, we successfully apply the low-cost LBL-SA method with polyelectrolytes as molecular connectors to fabricate a highly sensitive AuNPs/poly(allylamine hydrochloride) (PAH) flexible strain sensor. We proved the AuNPs/PAH strain sensor could detect eye-ball movements, facial expressions, and other motion and physiological signals such as finger bending and wrist pulses. Combined with machine learning, we successfully control the behaviors and emotional states (via controlling facial expressions) of animated characters in the virtual world.

2. Results and Discussion

2.1. Sensor Fabrication and Characteristic

Continuous adsorption of polyanion (citric acid anion) and polycation was used to create (AuNPs/PAH)_n (*n* is the number of layers) films from AuNPs solution and the cross-linking agent PAH (cationic poly(allylamine hydrochloride)). The electrostatic interaction between oppositely charged molecules drives the creation of structures.^[29] The fabrication process of the AuNPs/PAH film is illustrated in Figure 1a. Because of electrostatic repulsion between adjacent AuNPs in the same layer and electrostatic adsorption, the adsorption of AuNPs in films is dispersed and discontinuous (Single layer AuNPs is shown in Figure S1, Supporting Information). We have found that (AuNPs/PAH)₁₀ film has

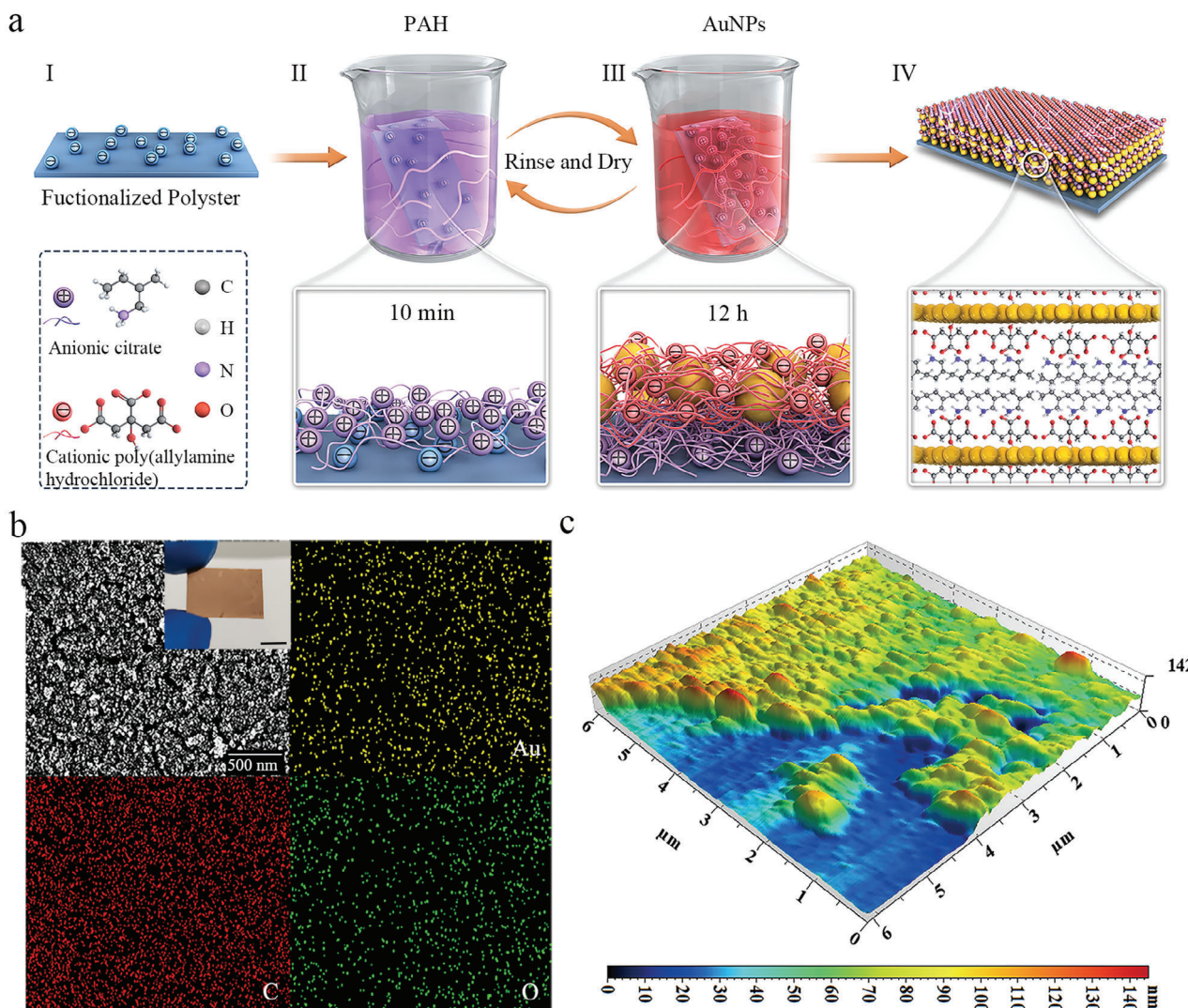


Figure 1. The schematic and characteristics of AuNP/PAHs films. a) The schematic of the preparation of AuNP/PAHs films. The negatively charged substrate (I, piranha solution modified) was immersed in cationic solution PAH (pH 4.5) for 10 min (II). After rinsing and drying, the substrate was immersed in an anionic AuNPs solution (pH = 6.0, and $\approx 10^{11}$ particles ml^{-1}) for 12 h (III). Repeat steps II to III until the target number of layers is achieved (IV). b) FIB (Focused ion beam) images and EDS mapping of the $(\text{AuNPs/PAH})_{10}$ film, characterizing the surface morphology and the distribution of the main elements of the film. (The inset is a physical drawing of the sensor, and the scalebar is 0.5 cm). c) The AFM image showing the surface morphology structure and height of the $(\text{AuNPs/PAH})_{10}$ film.

many gaps between the nanoparticles, which will provide a path for charge transfer within the film as it deforms (Figure 1b). The AFM image shows the overall topography of the nanocomposite, i.e., the height and morphology of the $(\text{AuNPs/PAH})_{10}$ film (Figure 1c). The uneven surface morphology indicates the random nature of AuNPs self-assembly. The obtained roughness of the surface is relatively low for metallic nanoparticles/polymer nanocomposite. The highest heights are at 140 nm, which remains reasonably flat for this kind of nanocomposite.

2.2. Sensing Mechanism and Properties

Because the AuNPs film is very thin (≈ 140 nm) and adheres well to the substrate, we consider that the stress distribution

of the substrate can represent the stress magnitude and direction of the AuNPs film under certain conditions. The strain sensitivity of the $(\text{AuNPs/PAH})_n$ films was measured with the bending of the substrates. The diagram of the test method is shown in Figure 2a. As the AuNPs film changes from initial deformation to tensile strain, the distance between two adjacent nanoparticles increases, widening the gap for charge transfer and leading to an increase in resistance. On the contrary, the gap between the nanoparticles will be reduced under compressive strain, which shortens the charge transport path and leads to a rapid and significant decrease in resistance. Since the thickness of the AuNPs/PAH film is on the nanometer scale, we assume that the stress on the substrate (PET) is completely equivalent to the stress on the film. The Finite-Element software COMSOL was used to simulate the stress distribution in the PET film material.

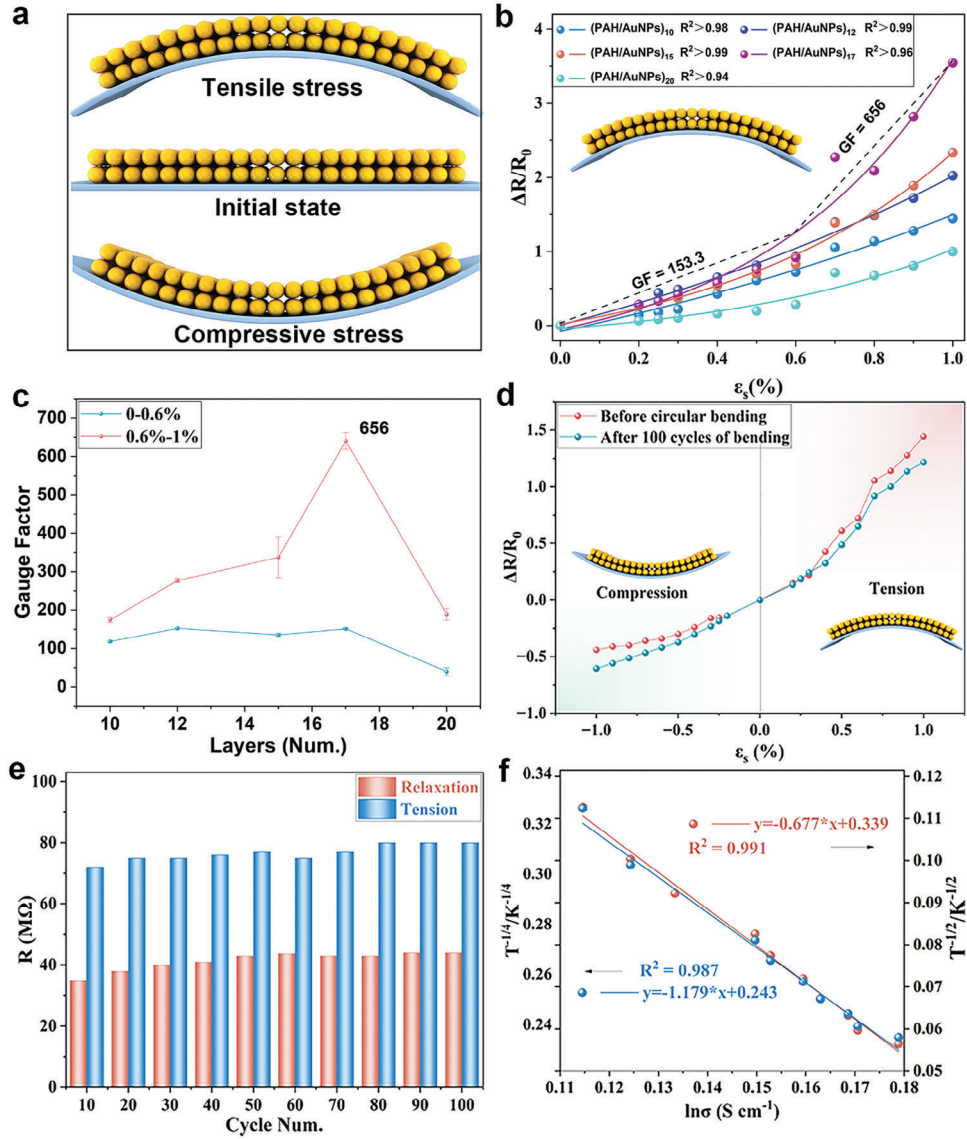


Figure 2. Results of the sensor's characteristics. a) Schematic of the flexible sensor under tensile and compressive stress. b) $\Delta R/R_0$ of (AuNPs/PAH)_n films under tensile strain and results of simulation Equation 2 ($n = 10, 12, 15, 17$, and 20). The dashed line represents the gauge factor of (AuNPs/PAH)₁₇. c) Comparison of the sensitivity of (AuNPs/PAH)_n films in the range of 0–0.6% and 0.6–1%, respectively. d) Comparison of the resistance of the (AuNPs/PAH)₁₀ film after repeated bending 100 times at 1% tension. e) (AuNPs/PAH)₁₀ film under tension and compression; the results of $\Delta R/R_0$ comparison before and after 100 bends were repeated. f) Arrhenius-like plots and linear fitting (solid line) of conductance.

The stress distribution in the cross-section and on the surface of the substrate at 1% stress is shown in Figure S2 (Supporting Information). The force at 1% stress was measured ≈ 0.9 N. Therefore, we calculated that Young's modulus of the AuNPs/PAH film is 300 MPa and the bending rigidity is 0.225 mN. The strain sensing performance of the AuNPs/PAH film sensors is evaluated by measuring the relevant resistance change as a function of uniaxial imposed strain. The thin films of layered structures composed of 0D AuNPs give them the transmission capability of unique channels. In this structure, the total charge transfer is primarily controlled by percolation theory and tunneling effects, which are very sensitive to the thickness of the dielectric layer (i.e., pure single PAH layer). Notably, the device exhibits ultra-high sensitivity

when the thickness of the PAH layer changes slightly under external stress.^[30] Therefore, the sensitivity of the different layers of AuNPs/PAH films varies considerably (Figure 2b). Considering the thickness of the PET substrate and Young's modulus, when the film strain reaches 0.7%, the radius of curvature of the cylindrical support is much smaller than before, and a larger stress must be applied to meet the experimental requirements, resulting in a sudden increase in the film bending radius and a sudden increase in the resistance change. It is further shown that our sensor is capable of responding not only to gradual increases in stress but also to sudden increases in strain promptly. After the substrate adapts to the bending arc, the resistance change gradually increases with the strain. Thus, we divided the sensitivity

(GF) of the sensor into two parts for calculation (i.e., 0–0.6% and 0.6–1%).

The electrical properties of conductive polymer composite mainly depend on the volume fraction of the filler.^[31] According to the prediction of percolation theory,^[32] we tentatively determined the percolation threshold (φ_c) of conductive polymer composite. Zhang et al. have demonstrated that the GF increases with increasing gold nanoparticle layers when the volume fraction of AuNPs is less than the threshold ($\varphi < \varphi_c$), and GF decreases with increasing gold nanoparticle layers when the volume fraction of AuNPs is greater than the threshold ($\varphi > \varphi_c$). The GF was maximum around the threshold, which is consistent with our results. In (AuNPs/PAH)_n films, an increase in *n* implies an increase in the volume fraction of AuNPs. It is reasonable because when the volume fraction of AuNPs is larger than the threshold ($\varphi > \varphi_c$), the particle spacing gradually decreases, particle agglomeration increases, and the electron transport mechanism changes from the tunneling effect to direct metal contact. In this situation, the conductivity of the AuNPs/PAH film will be higher, while the sensitivity will be lower. The high GF value is based on the fact that certain metal contacts act as conductive pathways near the percolation threshold (φ). At this point, the applied strain can alter the conductive pathway more effectively by breaking some of the remaining metal contacts. This is a key factor for achieving GF adjustability and high sensitivity.^[33]

In general, the resistance of a network composed of nanoparticles can be divided into two states: 1) a completely insulated state when all nanoparticles are widely spaced from each other and 2) a state when some nanoparticles are closely packed and behave like a monolithic block of metal, ideally without contact resistance.^[34] However, the organic matter in the ligand-bound nanoparticle film is critical in terms of electronic coupling between adjacent nanoparticles. Duan et al. attributed the electrical conductivity of nanoparticle networks to four factors: particle size, core-core gap, number of nearest neighbors, and the dielectric constant of the ligand molecules.^[35] Most nanoparticle-based strain sensors are based on nanofilm network structures with alkanes as molecular ligands.^[36] Common theoretical models for the electromechanical properties of nanoparticle thin film sensors are thermally activated carrier tunneling,^[37] and F-N tunneling effect,^[38] effective medium theory,^[39] molecular length junction resistance formula,^[40] etc. Here, we simulated the change in resistance of the film from the temperature-dependent Coulomb charging energy equation of the Arrhenius fitting for the nanoparticle array. The conductivity of the film can be described by a thermally activated tunneling model:^[41]

$$\sigma = \sigma_0 \exp(-\beta\delta) \exp\left(-\frac{Ea}{kT}\right) \quad (1)$$

According to the Coulomb model, the corresponding resistance change equation as a function of the strain coefficient (ϵ_s):

$$\frac{\Delta R}{R_0} = \exp(A\epsilon_s) \exp\left(B - \left(\frac{C}{9.8 + 18.9\epsilon_s}\right)\right) - 1 \quad (2)$$

Here, *A* is the constant related to β . *B* and *C* are the constants related to ϵ . The detain parameter is shown in Table S1 (Supporting Information). The results of the simulation are also shown

in Figure 2b. Furthermore, we analyzed the reasons for the errors between simulations and experiments:^[42] (1) The response of the 3D film structure to mechanical stress is a rearrangement of the particle structure, increasing the number of nearest neighbors of the nanoparticles (i.e., the increase in the charge transport path). (2) The involvement of PAH linker molecules in the charge transport mechanism reduces the sensitivity of the resistance to strain. (3) The morphological differences in the films affect the charge transport mechanism and the sensitivity. (4) The effect of nanoparticle aggregation on charge transport and sensitivity (Figure S3, Supporting Information).

The strain sensor of the (AuNPs/PAH)₁₇ film exhibits the highest sensitivity with a Gauge Factor (GF) of ≈ 656 (0.6–1%) (Figure 2c). In detail, it is shown in Figure S4 (Supporting Information). The results show that the film has the best repeatability and reproducibility when the *n* value is 17. However, the preparation time of 17-layer films is considerably longer compared to 10-layer films, while the 10-layer films also have high GF (180, 0.6–1%). Considering the efficiency and sensitivity, the 10-layer film sensors were selected for application development, and the following experiments were conducted using sensors with 10-layer films. A series of typical data showing the relevant increase/decrease in resistance with increasing tensile/compressive strain is presented in Figure 2d. It exhibited the repeatability of the sensor by measuring the resistance changes after 100 cycles of bending. The results revealed that the GF can still reach 141.9 (0.6–1%). In contrast to compression, we considered the tensile test to be more reflective of the recovery performance of the sensor, so we tested the cycling results at the maximum tensile strain condition. The concrete change in resistance induced by repeated convex bending to the substrate is shown in Figure 2e, confirming the excellent reproducibility and reversibility of the response from the strain sensor within a tension of 1%. It is worth mentioning that in different layers of AuNPs/PAH sensors, the surface structure is also different. The surface of films with higher sensitivity from (AuNPs/PAH)_n (*n* = 12, 15, and 17) films shows nano-cracks (Figure S5, Supporting information), and the (AuNPs/PAH)₂₀ film is more stable than other layers. The existence of nano-cracks in metal/elastomer nanocomposites has been assigned to a mechanical incompatibility between the two components of the nanocomposite due to differences in the Young modulus and the thermal expansion coefficient.^[43] However, the nano-cracks are usually very narrow and keep a close enough proximity between the metal nanoparticles to allow a conductive path. The effect of the nano-cracks on the resistance can be intuitively understood as the nano-cracks in the AuNP/PAHs network would block the charge transfer, thus increasing the resistance of the sensor. It can be reasonably stated that the AuNPs layer based on the nano-cracking process adds value to the sensitivity of the thin film sensor since a larger and very sudden resistance change suggests a higher GF value, which is compatible with the experimentally obtained values of the GF.

Metallic nanoparticles/polymer films with nanoparticle volume fractions below the percolation threshold generally have high resistance, but under strain conditions, the nanoparticle distributions in the film become more dispersed or tighter, which increases or decreases the resistance of the film.^[44] These piezoresistive sensors with metal nanoparticles and polymer substrates are relatively new members of the flexible sensor family, and

they typically use the quantum tunneling model and percolation theory to describe the sensing mechanism.^[45] The device construction that most closely resembles the electron tunneling concept, in which nanoparticles are joined by flexible molecular linkers,^[46] is shown to produce the highest sensitivity result from the ultrathin film. Because the conditions for triggering high-sensitivity (AuNPs/PAH)_n thin-film strain sensors are compression and tension, our sensors can be widely used for motion and health monitoring. In addition, the temperature dependence of the (AuNPs/PAH)_n films brings more complexity to the conduction mechanism and more interesting situations to consider when designing the sensor. In general, the temperature constraints appear challenging in sustaining the electrical behavior for a large number of nanodevices. The conductivity of (AuNPs/PAH)_n films is typically described by the thermal-activated tunneling effect of the carrier. The charge injection and transport, as well as the trapping/de-trapping dynamics of carriers in such metal nanoparticles/polymer nanocomposites, are controlled by both the ability of the metal nanoparticles to store electrical charges and the properties of the surrounding matrix, in terms of the density of charge trapping centers and their energy level distribution.^[47]

Herein, we studied the temperature dependence of the conductivity and resistance of the (AuNPs/PAH)₁₀ film in the low-temperature range (80 to 300 K). Thermally activated diffusion of metal atoms and/or very small metal nanoparticles is not considered in the analysis since the diffusion length of the latter is less than a nanometer for these low temperatures. The polyelectrolyte PAH solution and the citrate ions encapsulated AuNPs will provide electrons, and ions in the polyelectrolyte that will tunnel through the gap between adjacent AuNPs to generate the tunneling current.^[48] The tunneling of electric charge into the gap between two AuNPs will generate Coulomb energy by a charging effect. And it turns out that by combining the tunneling effects and Coulomb charging energy of the metal nanoparticles, the non-metallic (dielectric type) conduction behavior may be explained.^[49] We investigated the (AuNPs/PAH)₁₀ sensor conductivity concerning the temperature (Figure S6, Supporting information). In the range of 80 to 300 K, we measured the Arrhenius-like plots of the change in conductivity of the sensor as shown in Figure 2f. The conductivity conforms to a linear dependence. For the influence of temperature on the conductivity, due to the random distribution of AuNPs on the substrate, two main transfer models, VRH and TMM, can be considered at the origin of the sensing mechanism of the AuNPs/PAH film.^[50] Therefore, the electrical transmission characteristics of the (AuNPs/PAH)₁₀ films can now be understood based on the mutual action of both conduction modes: percolation model for the variable range hopping conduction model and tunneling mechanism conduction model for charge transport between two adjacent AuNPs. We thus provide a reliable criterion for revealing the conductive mechanism of the (AuNPs/PAH)_n films.

2.3. Physical Behavior and Micro-Activity Monitoring of the Human Body

Human joint and muscle movements are essential to maintain human activity. For patients in recovery, the detection of limb

joint movement, muscle movement, and pulse signals are very important because they can provide important medical data for the targeted treatment of human diseases.^[51] Therefore, it is crucial to monitor and record the information about human movement in real-time for studying the recovery state of patients, and it has a broad application prospect in the medical field.^[52] We assess our sensors by monitoring human limb movements, including finger flexing, wrist pulsation, and upper arm muscle extension in Figure 3. The sensor is attached to the front or rear of the finger, and we measure the signal change caused by the compression or tension (Figure 3a). Additionally, our sensors can detect the muscle activity on the arm during fist clenching (tense or relaxed) (Figure 3b). Wrist pulse signals are very important, they can predict human blood pressure, heart rate, etc.^[53] The sensor can be applied to detect wrist pulses in real time under relaxed conditions (Figure 3c). Three different characteristic peaks, namely shock wave (PW), tidal wave (TW), and diastolic wave (DW), can be identified from the measured pulse profile shown in Figure 3d (the dashed box part in Figure 3c). These characteristic peaks are not found in previously reported pressure transducer measurements.^[54] The ratio of PW and TW waves (i.e., the augmentation index of the radial artery $AI = TW/PW$) is important information for determining arterial stiffness as well as blood pressure,^[55] while the time difference (ΔT_{DVP}) also provides an important parameter for cardiac health. This further demonstrates the application of this sensor in the monitoring of human physiological signals.

2.4. Monitoring of Human Eye-Ball Activities and Emotional State

Perceiving human facial emotion and eye movement is also an important part of advancing technologies in personal communication, entertainment, and security systems. Eye-tracking helps to increase the user's sense of immersion in VR. Viviane et al.^[56] used VR systems with a motion tracker and built-in eye tracker and showed the results obtained by using eye-tracking in VR. They constructed a 3D gaze vector from the 2D pupil position and then intersected the vector with objects in a 3D world. It can record the number and time of houses visited by subjects. Moreover, Yun et al.^[57] use eyes and surface electromyogram (EMG) as novel input methods of VR to provide a portable and immersive virtual experience. VR devices with eye-tracking technology also open many possibilities for real-time attention assessment.^[58] Eye tracking, as an input device in the metaverse, plays an important role in immersive experiences.^[59] Here, we explored the usage of the developed high-sensitivity sensor to detect different eye movements. The sensor was placed on appropriate locations on the face to detect eye movement and blinking, as shown in Figure 4. Eye movement detection refers to the perception of the muscles involved in eye movement and blinking. Figure 4a–d shows the $\Delta R/R_0$ response of a sensor attached to the skin beneath the right eye to the muscle group around the eye when the subject's eye looked up, down, left, and right. The $\Delta R/R_0$ of the sensor also gave a corresponding feedback signal when the subject's eyes were rotated in all directions from the middle and then returned to the original position. Likewise, when the sensor was fixed at the same position, Figure 4e and f show the feedback signals of blinking naturally

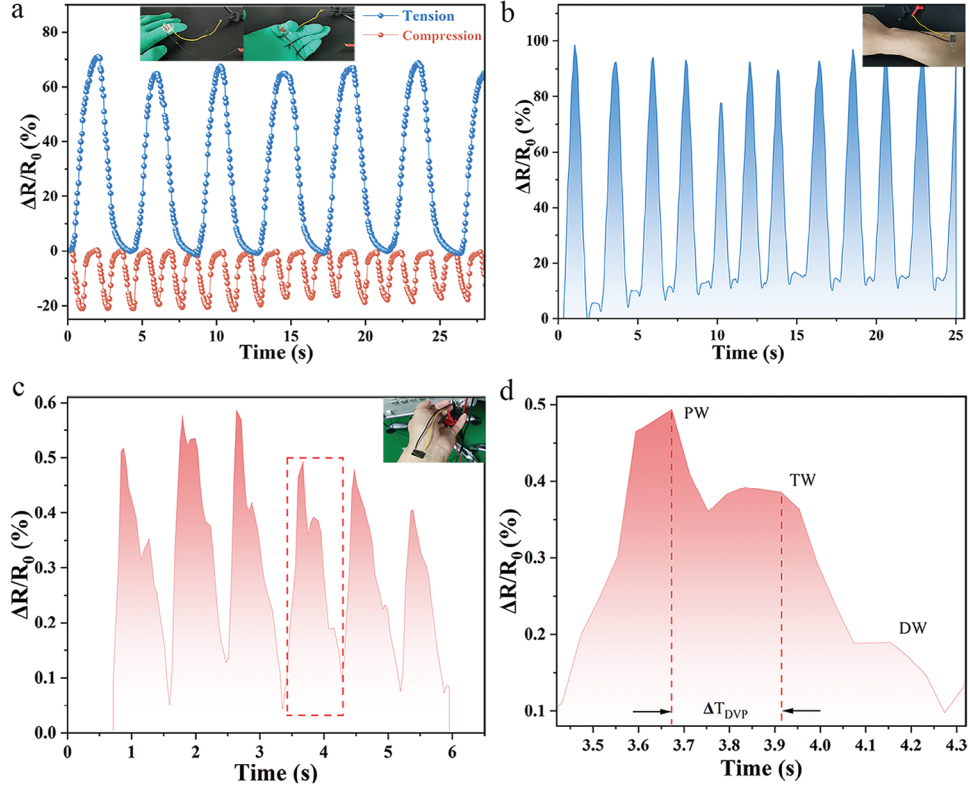


Figure 3. Monitoring of human limb behavior and muscle movement. a) The output signals of bending a finger. By placing the sensor in different positions of the finger, the sensor is subjected to tensile or compressive stress. b) The signal of arm muscle movement. First put the sensor on the arm, through the hand-clenched fist-relax cycle movement, when the arm muscles are in a state of expansion-relaxation. It exerts tension on the sensor so that it is subjected to tensile strain. c) By attaching a sensor to the wrist, which can monitor the wrist pulse signal of a normal human. d) An enlarged version of the pulse signal in the dashed box in Figure 3c.

and winking (squeezing the eye). Although both times the eyes are closed, squeezing the eyes is conscious, involving more intense muscles around the eyes and moving at a greater range, leaving the sensor in a bent state with a resistance change of up to 9%. Blinking, on the other hand, is involuntary and involves relatively small muscle movements. The results of the sensor's response to both phenomena can be seen clearly in the image.

On the other hand, the immersive and interactive experience of virtual technology provides the possibility to reveal the mechanism of emotional cognition.^[60] Researchers have previously thought that there are six different identifiable facial expressions: *happiness*, *sadness*, *fear*, *anger*, *surprise*, and *disgust*. However, a recent study^[61] from the University of Glasgow suggests that some of these share facial “signals” and should be combined. For example, they argue that fear and disgust share the same key facial signal, i.e., the wrinkled nose. The same goes for fear and surprise, i.e., both involve widened eyes and should be combined. Since there are 42 individual facial muscles in the face, to examine how each of these muscles impacts facial expressions, researchers typically work with specially-trained subjects who are able to activate every one of the 42 muscles. Thus, a sufficient number of facial muscles are studied to allow observers to distinguish between different emotional states through facial expressions. The complex facial expressions of human beings

can present corresponding psychological and emotional changes. Therefore, to study the performance of our sensors as an “emotion detector”, i.e., extending it to monitor subtle changes in human facial expressions, we have attached the sensors to the skin around the eyes and corners of the mouth and observed the sensors' output to the change in the volunteers' facial expressions. Generally speaking, when a person presents different emotional states, it drives the corresponding muscle groups, of which the angular puller and the labial presser are the main facial muscle groups involved in different facial expressions.^[62] Consequently, we chose to mount the sensors at the corners of the volunteer's mouth to detect different expression signals. Here, we measured six human expressions: *happiness*, *surprise*, *sadness*, and *disgust*, which are shown in Figure 4g–j. From the change in resistance ($\Delta R/R_0$), we can see the different feedback signals for different facial expressions. This indicates that our sensor has the capacity to recognize human emotional states. Further combination of machine learning to classify and recognize different expressions, the classification result is shown as the confusion matrix in Figure 4k, and the macro-recall of the four expressions is 84.5%. We further tested the sensor on more human subjects for facial expression recognition. Figure 4l shows the classification results of ten subjects in total (No.1–No.10). The mean recall values for expressions of happy, surprise, sad, and disgust are 82.2%, 85.6%, 87.1%, and 86.3%, respectively. In addition, the

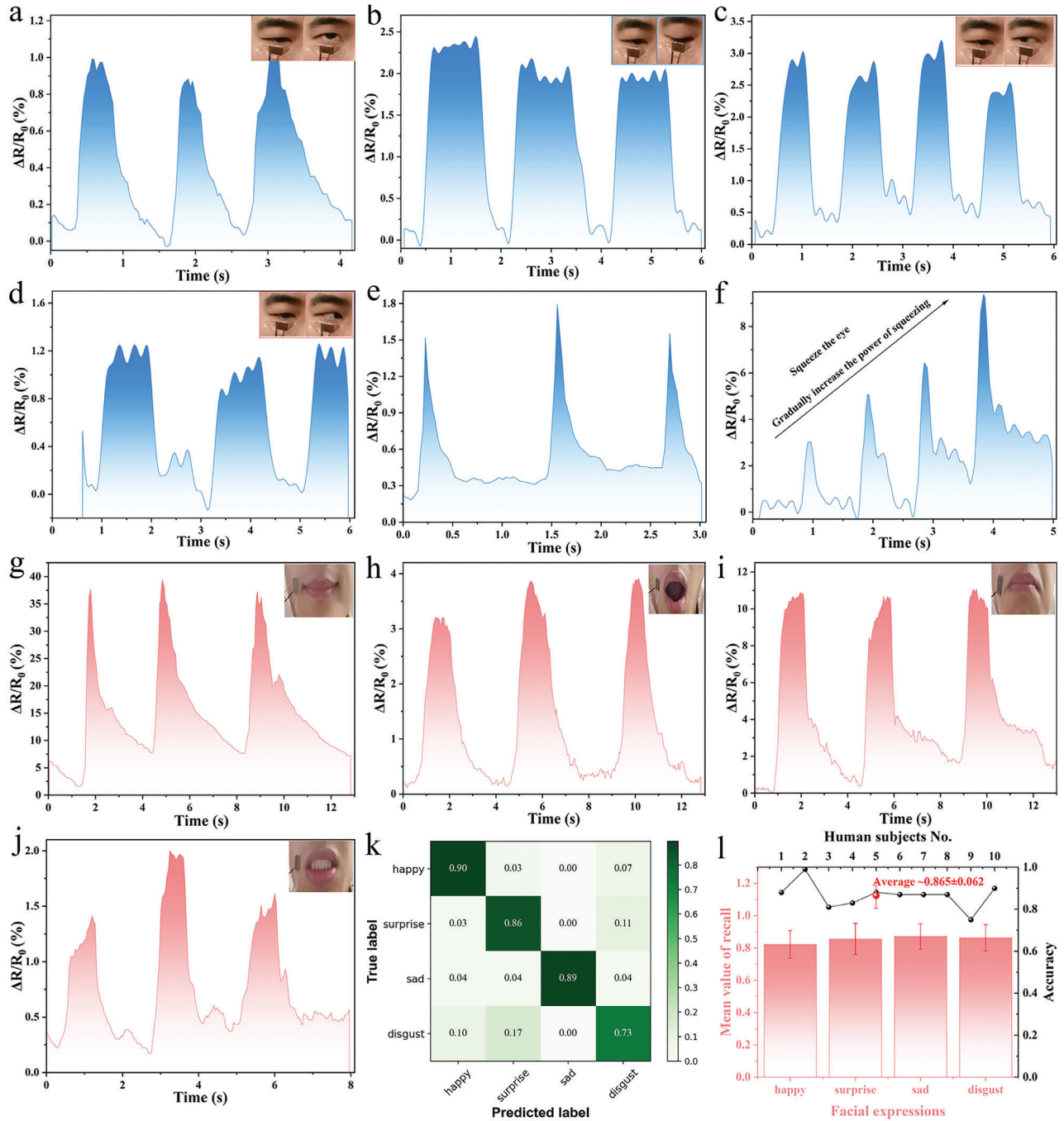


Figure 4. Results of the eyeball movements and facial expressions detection. a–d) represent the variation of responses of the sensor attached underside of the eye when the volunteers looked up, down, right, and left, respectively. e) and f) blinking and winking (here, blinking means closing and opening eyes normally; winking means a conscious way to squeeze the eyes. Gradually increase the power of squeezing, and its signal changes correspondingly). g–j) show the signal feedback of different facial expressions (in the order of happiness, surprise, sad, and disgust). All signal widths were correlated with the frequency and the duration of the dwell point of eye movements and expressions. k) The confusion matrix of four facial expressions. l) The individual recognition accuracy of facial expressions from ten subjects.

average recognition accuracy of the subjects is $\approx(86.5 \pm 6.2)\%$. The classification matrixes of the other nine subjects are shown in Figure S7 (Supporting Information). When two sensors are applied, i.e., adding one sensor at the eyelids (the experimental schematic of two sensors is shown in Figure S8, Supporting

Information), six facial expressions (i.e., *happiness*, *surprise*, *sadness*, *anger*, *disgust*, and *fear*) can be recognized with an average accuracy of $\approx(97.3 \pm 2.9)\%$ for three human subjects. The confusion matrixes are shown in Figure S9 (Supporting Information).

2.5. Controlling an Avatar's Behaviors and Facial Expressions

The rapid development of virtual reality technology provides ideas for the metaverse to become a diverse application in social, entertainment, medical, gaming, and work areas.^[35] To exhibit the application of our sensors in the metaverse, we demonstrate controlling a virtual avatar's activities (i.e., walking in different directions and facial expressions) with the detected signals of a real human subject's activities (i.e., eye movements in different directions and different facial expressions). First, we used one sensor (sensor #1) to detect eye movements (i.e., *look-up*, *look-down*, *look-right*, and *look-left*), and the schematics are shown in the basic process for the animation control system in Figure 5a. Then, a random forest algorithm was applied to build the classification models with the datasets to predict the specific motion. Five-fold cross-validation results are shown in Figure S10 (Supporting Information). The confusion matrix of eye movements is shown in Figure 5b, and the macro-recall reaches $\approx 91.5\%$. Finally, we further employed two sensors (sensor #1 and sensor #2 in Figure 5a). The confusion matrix for the facial expression datasets is shown in Figure 5c, and the macro-recall reaches $\approx 98.8\%$.

With the highly accurate recognition, the testing data (here, the data is the same as the Figure 5b,c) of multi-activities were fed into the trained models, and the corresponding predicted motion commands were then sent to the virtual space built with Unity for animation control. The demonstration of VR videos (Video S1 and Video S2, Supporting information) shows the great potential of our sensors to achieve advanced multi-functional as well as diverse applications, providing ideas for exploring the development of the metaverse. Figure 5d shows the signals of human activities and the corresponding animation characters' activities. The virtual character will follow the specified movements controlled by humans in physical space. For example, when we look up, the avatar will move forward. Similarly, when we perform different human emotional expressions, the avatars will replicate our actions, which provides a good foundation for further implementation of virtual character emotionality in the metaverse. In the future, intelligent systems based on flexible electronic skins could further human-metaverse integration.

3. Conclusion

We developed a cost-effective and highly sensitive mechanical strain sensor based on an AuNPs/PAH composite film, which is built using a layer-by-layer self-assembly process by absorption of positive and negative charges on flexible substrates. Our experimental results demonstrate the nanoparticle-based strain sensor's reversibility, reproducibility, and excellent sensitivity. Moreover, the strain sensor could monitor facial expressions and eye movements. Combined with machine learning and virtual reality technologies, the sensor could control the virtual avatar's movements and emotional expressions through the movement of the eyeball and human facial expressions in physical space. These findings demonstrate that our sensors can be extensively implemented in interfacing with the metaverse space. Potentially, the sensor can also be applied as a wearable health monitoring device to provide real-time status of a patient's emotional state, i.e., *sad*, *happy*. To further predict the patient's negative emotions due

to illness, such as depression, pain, etc., so that they can receive better treatment and care from doctors and family members.

4. Experimental Section

Materials and Equipment: Chloroauric acid (HAuCl_4), sodium citrate, and Poly (allylamine hydrochloride) (PAH, Mw = 17 500) were purchased from Sigma-Aldrich. All the chemicals were applied without any further process. Ultrapure water (resistivity $18.2 \text{ M}\Omega\cdot\text{cm}$) was used in all the experiments. The model of polyester film was HD-PET and the brand was DuPont. The Helios G4 CX field emission high-resolution Focused Ion Beam (FIB) and The Agilent 5500 AFM (Agilent Technologies Company, USA) were employed for characterization. Tensiometers (HANDPI) were used to measure the stress of the sensor at a strain of 1%. In all experiments, the sensor was attached by the 3 M Tegaderm film.

Synthesis of AuNPs: The preparation of AuNPs was performed according to the standard reduction method of sodium citrate.^[29] In brief, sodium citrate (2 ml, wt. 1%) aqueous solution was added to the boiling HAuCl_4 (100 ml 1 mmol) solution. The mixed solution was stirred and heated in a magnetic stirring device for 10 min, then the heat source was removed, and the mixture continued to be stirred until it cooled. Finally, the solution was wine-red color, and the average diameter of AuNPs was $\approx 18 \pm 3 \text{ nm}$. After the synthesis of the solution, sodium citrate covered the surface of AuNPs as a stable group. These AuNPs were moderately negatively charged under normal pH conditions.

Preparation of LBL-SA films for AuNPs: A 10 mm $\times$ 15 mm Polyester film with a thickness of 0.2 mm was used as the flexible substrate. First cleaned the PET in fresh Piranha solution (3:1(v/v) $\text{H}_2\text{SO}_4/\text{H}_2\text{O}_2$) for 30 min, then rinsed thoroughly with plenty of ultrapure water to dry. Finally, a substrate was obtained with negative charges. The AuNP/PAH film was fabricated by the LBL-SA deposition. Since AuNPs were prepared from sodium citrate, they were coated with negatively charged citrate. First, to make the AuNPs adhere to the PET, a layer of positive charge, choosing a polyelectrolyte solution of PAH was deposited. Second, the substrate was immersed into the PAH (0.1%) solution for 10 min, then rinsed and dried with ultrapure water. Eventually, the PET was placed in AuNPs solution for 12 hours and then rinsed twice with ultrapure water. The membrane was finished until the desired number of layers was achieved.

Resistance Measurements: To detect strain-induced resistance changes the devices were placed into sample holders allowing for defined convex and concave curvature bending of the substrates.^[41] The strain factor ε of the film could be calculated using the equation: $\varepsilon = \pm L/(2R \pm L)$. Here, L is the thickness of the substrate (0.2 mm) and R is the radius of curvature of the sample holders. The strain factors ε were 0, 0.2%, 0.25%, 0.3%, 0.4%, 0.5%, 0.6%, 0.7%, 0.8%, 0.9%, and 1%. The direction of bending was always subject to the direction of curvature bending. A Keithley digital multimeter and Oscilloscope (5 Series MSO54, Tektronix) were applied to collect the signals of resistance and voltage from sensors.

Experiments on Human Activity Detection: *Finger-bending experiments:* Place the sensor on the front or back of the finger and detect the change in the resistance signal generated when the finger bends.

Arm-muscle experiments: Place the sensor on the upper arm and detect the change in resistance signal when the fist clenches (tense or relaxed).

Eye-movement experiments: One sensor was attached to the subject's skin surface at the round lower eyelid of the right eye with the transparent film dressing (3 M Tegaderm), and the human subject performed the motions of moving the eyeball in four different directions, i.e., *look up*, *look down*, *look left*, and *look right*, with each direction repeating for 120 times.

Facial expressions experiments: First, one sensor was attached to the subject's corner of the mouth, and ten subjects performed four facial expressions, i.e., *happiness*, *surprise*, *sadness*, and *disgust*, respectively. Ten subjects were recruited, and each subject was asked to repeat 20 times for each of the four expressions. Then three more human subjects were asked to perform six different expressions of happiness, surprise, sadness, anger, disgust, and fear by wearing two sensors, i.e., one was around the lower eyelid and the other was around the corner of the mouth. Each expression was repeated 30 times for each subject. All the activities were performed

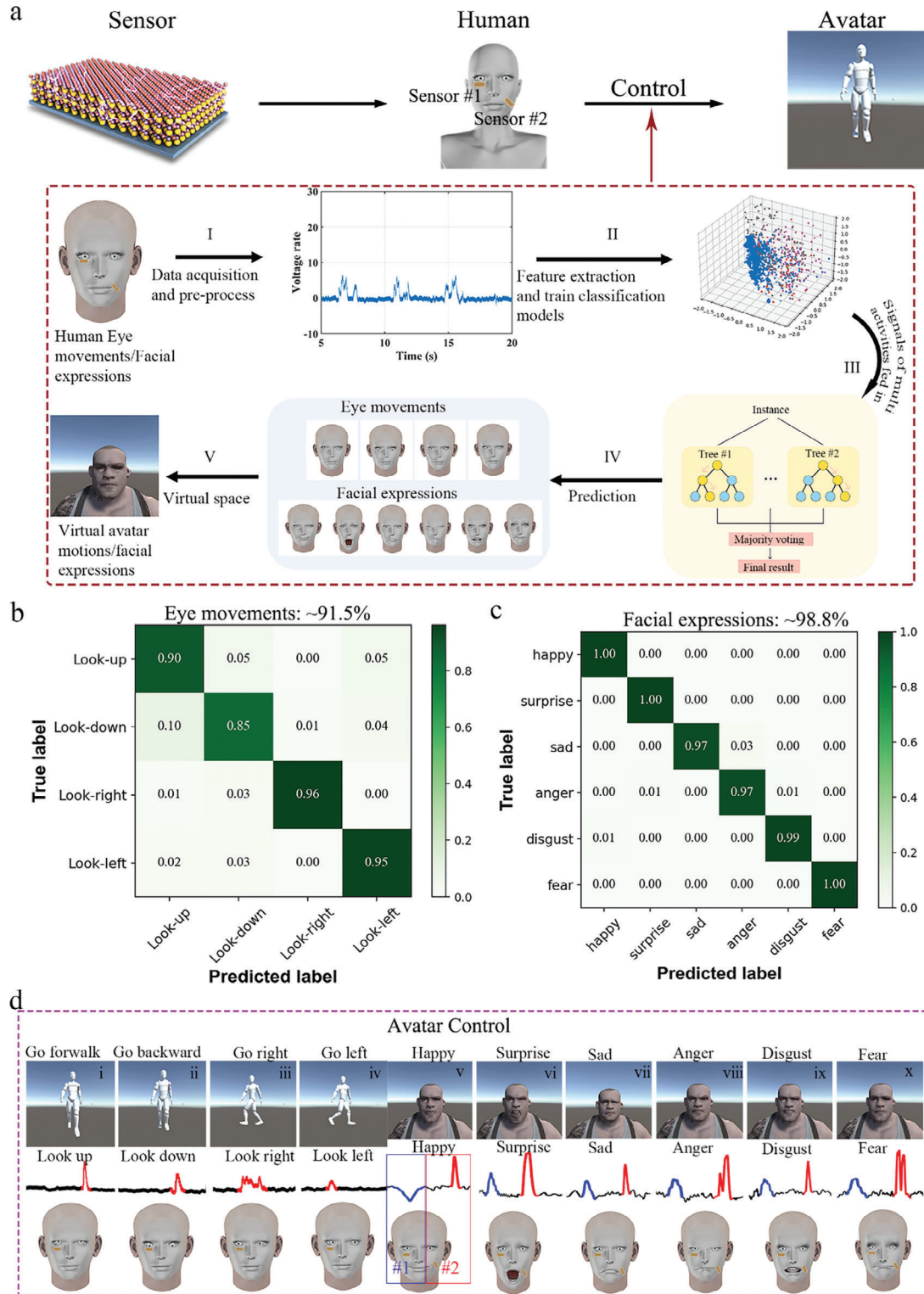


Figure 5. Nanoparticles-based sensors combined with machine learning for human eye movements, facial expression recognition, and the control of virtual characters. a) Controlling animation character process. I) Acquire the data from humans; II) low pass filtered, fixed windows segmentation, feature extraction, and classification (random forest algorithm); III) feeding a series of signals of different motions into the trained classification model; IV) prediction motion; V) realizing virtual avatar controlling by Unity. b) and c) Confusion maps of eye movements and facial expression recognition derived from the machine learning model. The true label refers to the recognized results, and the predicted level refers to the physical objects. d) The behavior and facial expressions of virtual avatars corresponding to real human actions in physical space. The blue data (Figure 5d,v-xdown) indicates the data from sensor #1, and the red data curve indicates the data from sensor #2.

every four seconds and the signals were sampled by oscilloscope and multimeter (5 Series MS054, Tektronix, and Keithley DMM 7510 Bench 7.5 Digit Multimeter).

Data Acquisition and Processing for Animation Control: Signals of human motions of eye movements and facial expressions were demonstrated for animation control. As the process shown in Figure S11 (Supporting Information), all the signals were first low-pass filtered at 5 Hz (to remove electromagnetic noise) and the rates of changes were calculated. Then, the data were segmented into a fixed window of four seconds. The time-domain average, root mean square, standard deviation, and cumulative length of each window were calculated (Equation S7–S10, Supporting Information) as part of the feature matrix. In addition, the number of peaks in each window was also extracted. The recursive feature elimination with cross-validation algorithm (RFECV) was used to find the optimal number of features, and then the random forest algorithm was used for classification. Five-fold cross-validation was used to make better use of the data. The metrics, recall, macro-recall, and accuracy were calculated as expressed by Supporting Equation S11–S12 (Supporting Information) to evaluate the classification performance of each class and the overall performance. The animation controlling process was realized by feeding a series of signals of different motions into the trained classification models, and then the predicted motion was sent to Unity for animation control.

Statistics: All data shown in this manuscript were representative. The number of the data in Figure 2c was given as mean and standard deviation. The processing methods of the data were offered in the subsection “Data acquisition and processing for animation control” and Figure S11 (Supporting Information). The sample size for the human test and calculated methods of the confusion matrices were provided in the subsection “Experiments on human activity detection”. Statistical analysis was carried out using OriginPro, MATLAB, and Python.

Ethics Approval Statement: Human and Artefacts Ethics Subcommittee, City University of Hong Kong gave ethical approval for the human test in this work (Reference No: HU-STA-00000215).

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Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

J.W. and J.S. contributed equally to this work. W.J.L. and Z.W. supervised and guided the project; W.J.L. X.Y., and Z.W. conceived the initial concept design; H.Z. investigated the initial background review; J.W., J.S., H.Z., and M.G. performed the experimental design; H.Z. collected the data of eye movement and facial expressions; R.L., L.C., and K.W. helped to test the sensor performance; J.W. performed the analysis of the sensor performance; J.S. performed the data classification and controlled the animated character; R.V., K.M., and S.C.C. provided technical insights into material synthesis and fabrication. J.W., J.S., and H.Z. performed the drafting of the article; R.V., K.M., X.Y., S.C.C., W.J.L., and Z.W. performed the critical revision of the article for important intellectual content.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

avatar controlling, E-skin electronics, facial expressions, human eye tracking, nanoparticles-based sensor

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